Dr. Víctor Manuel Chávez Pérez



Professor–Researcher · Applied Physics, Data Science & Science Education

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Google Scholar · ResearchGate · LinkedIn
SNII · Candidate (SECIHTI) 2025–2028

PROFILE

I am a physicist from the University of Guadalajara and hold a PhD in Applied Physics from the University of Vigo. My research blends atmospheric and climate physics with large-scale data analysis: I study sudden stratospheric warmings (SSW), heat waves and their response under climate-change scenarios using models such as CMIP5 and WACCM.

Beyond basic research, I bring physics to real-world problems: I am developing a flood early-warning system for the Guadalajara Metropolitan Area that integrates weather radar, hydrological runoff modeling over high-resolution LiDAR terrain and remote sensing. My work also spans physical oceanography, remote sensing and applied optics.

I am also interested in how generative AI can transform science education: I design learning assistants and teaching materials that I evaluate with students themselves. Alongside this, I co-founded and coordinate the Tlakakini Science & Technology Club, devoted to science outreach for children. I believe the best science is the kind that gets shared.

RESEARCH INTERESTS

Sudden stratospheric warmings · Stratospheric dynamics · General circulation models (CMIP5 / WACCM) · Climate change · Physical oceanography & remote sensing · Applied optics · Data science · High-performance computing · Science education & outreach · Generative AI in science education · Machine Learning applied to climate & atmospheric data · Student perception of AI-based learning tools

EDUCATION

2017 – 2024 **PhD in Applied Physics**
Universidade de Vigo · Spain
Thesis: Sensitivity of sudden stratospheric warmings to simulation and analysis conditions in models.

2011 – 2012 **MSc in Climate Science: Meteorology, Physical Oceanography & Climate Change**
Universidade de Vigo · Spain
Temporal and spatial variability of the Minho river plume (MODIS 250 m). Top honors.

2009 – 2011 **MSc in Hydrometeorology — Meteorology & Physical Oceanography**
Universidad de Guadalajara · Mexico
Seasonality and anomalies of oceanographic variables in the Gulf of Mexico via remote sensing.

2000 – 2008 **BSc in Physics**
Universidad de Guadalajara · Mexico
Thesis: Ocean and coastal meteorology of Nayarit (San Blas, 1999–2002). Grade 95/100.

EXPERIENCE

2026 – present **Adjunct Professor ("Profesor de Cátedra")**
Tecnológico de Monterrey · Campus Guadalajara · Zapopan, Jalisco · Mexico
Course unit F1025B, The Laws of Motion and Conservation, team-taught by two instructors — one for theory and one for the challenge. Two theory groups and one challenge group.
Courses: The Laws of Motion and Conservation (Theory) ×2 · The Laws of Motion and Conservation (Challenge)

2024 – present **Lecturer "B"**
Centro Universitario de Tlaquepaque · UdeG · Tlaquepaque, Mexico
Department of Basic Sciences. Basic Sciences Academic Board: Secretary (2026 – present); Chair (2025).
Courses: Calculus ×2 · Advanced Calculus ×2 · Numerical Methods ×2 · Probability · Electromagnetism Lab · Discrete Mathematics

2024 – present **External Collaborating Researcher**
Universidade de Vigo · EPhysLab · Ourense, Spain
Doctoral research on sudden stratospheric warmings in CMIP5 and WACCM models; statistical analysis of large climate datasets.

2025 – 2026 **Lecturer**
Centro Universitario Bonpland y Humboldt · Online
PhD in Aerospace Engineering: Research Methodology and Fluid Mechanics.
Courses: Research Methodology ×2 · Fluid Mechanics

2019 – 2026 **Adjunct Professor ("Profesor de Asignatura B")**
Universidad Tecnológica de Jalisco · Guadalajara, Mexico
Mechatronics Division — Basic Sciences.
Courses: Engineering Physics ×10 · Probability & Statistics ×8 · Engineering Mathematics II ×6 · Differential Calculus ×5 · Integral Calculus ×3 · Engineering Mathematics I ×2 · Multivariable Calculus · Applied Engineering Statistics · Thermodynamics

2019 **Faculty Lecturer**
Tecmilenio · Campus Guadalajara · Guadalajara, Mexico
High school: Matter & Energy, Integral Calculus, Science & Technology.
Courses: Integral Calculus ×2 · Differential Calculus · Mathematics (Language of Science) · Matter & Energy · Science & Technology

2019 **Physics Instructor**
UNITEC · Campus Guadalajara · Guadalajara, Mexico
Engineering programs: Statics.
Courses: Statics

2015 **Predoctoral mobility stay**
Universidad Pablo de Olavide · Seville, Spain
Short research stay at R&D centers; results presented at international conferences.

2013 – 2017 **Predoctoral Researcher (FPI Fellowship)**
Universidade de Vigo · EPhysLab · Ourense, Spain
Statistical analysis of large-volume data for climate-change research with general circulation models.

2013 – 2014 **Predoctoral research stay**
Universidad de Guadalajara · Guadalajara, Mexico
Academic research stay on climate-change analysis.

2010 – 2012 **Physics & Math Instructor**
Centro de Enseñanza Técnica Industrial (CETI Colomos) · Guadalajara, Mexico
Engineering programs.
Courses: Differential & Integral Calculus ×2 · Statics ×2 · Dynamics ×2 · Mathematics ×2

2010 – 2011 **Physics & Math Instructor**
Universidad de Especialidades · Guadalajara, Mexico
Mechanics, Introduction to Physics and Electromagnetism.
Courses: Mechanics · Intro to Physics · Electromagnetism

2008 – 2011 **Professor–Researcher**
Instituto Tecnológico Superior de Zapopan · Zapopan, Mexico
Mathematics for engineering; parallel computing applied to oceanography.
Courses: Physics (Mechanics, E&M, Thermodynamics) ×11 · Mathematics & Calculus (Differential, Integral, Vector, ODEs) ×17 · Dynamics

2006 – 2007 **Research Assistant**
Universidad de Guadalajara · CUCEI, Departamento de Física · Guadalajara, Jalisco · Mexico
"Oceanografía de la Costa de Nayarit" project (CONACYT-SEP-2003-C02-44870F). PAY and PITAI programmes.

2005 – 2006 **Physics, Chemistry and Mathematics Teacher**
Centro Universitario UTEG · Guadalajara, Jalisco · Mexico
Intensive semi-schooled high school programme (BIS).

PUBLICATIONS

2025 **Observación Astronómica en México: Pasando de lo Visible a lo Invisible**
Almaguer-Gómez C., Medina J. F., Chávez-Pérez V. M., et al. · *Óptica Pura y Aplicada 58(1):15* · DOI: 10.7149/OPA.58.1.51200

2025 **Influence of Satellite and Presatellite Periods on the Validation of Major Sudden Stratospheric Warmings in Historical CMIP5 Simulations**
Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere 16(5)* · DOI: 10.3390/atmos16050628

2025 **Temporal Variability of Major Stratospheric Sudden Warmings in CMIP5 Climate Change Scenarios**
Chávez-Pérez V. M., et al. · *Climate 13(10)* · DOI: 10.3390/cli13100207

2023 **Estacionalidad y Anomalías del Nivel del Mar en el Caribe por Medio de Datos de Altimetría**
Palacios Hernández E., Chávez Pérez V. M., Sierra Romero L. L. · *Generis Publishing* · ISBN: 979-8-88676-642-4

2022 **Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Sudden Stratospheric Warmings**
Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere 13(4)* · DOI: 10.3390/atmos13040546

2018 **The climatology of dust events over the European continent using data of the BSC-DREAM8b model**
Mandija F., et al. · *Atmospheric Research 209:144–162* · DOI: 10.1016/j.atmosres.2018.03.006

In preparation

2026 Science Beyond the Classroom: Impact Assessment of a Monthly Science Club During National Non-School Days in Mexico
Almaguer-Gómez C., Chávez-Pérez V. M., et al.

2026 Student Perception of a Generative AI-Based Calculus Learning Assistant: A Mixed-Methods Study
Chávez-Pérez V. M., et al.

2026 Assessment of the lightning activity response to aerosol optical depth
Mandija F., Chávez-Pérez V. M.

CONFERENCES & TALKS

2026 **INNOVA FORUM 2026 Educational Innovation Meeting “Flexible Education and Digital Transformation” — Universidad de Guadalajara (UDG Virtual)**
10–11 sep 2026 · Guadalajara, Mexico
• AI-generated educational comics as a teaching strategy in higher education

2022 **65th Mexican National Physics Congress**
2–7 oct 2022 · Zacatecas, Mexico
• Analysis of the occurrence frequency of Southern Hemisphere stratospheric sudden warmings in high-top CMIP5 models
• Optical characterisation of transparent 3D-printing resin

2020 **63rd Mexican National Physics Congress**
4–9 oct 2020 · Online
• Significance of basic SSW characteristics when comparing CMIP5 models with reanalysis data over pre-satellite and post-satellite periods
• Impact on the occurrence frequency of major stratospheric sudden warmings under different climate change scenarios using CMIP5

2018 **American Geophysical Union (AGU) Fall Meeting 2018**
10–14 dic 2018 · Washington, USA
• Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Stratospheric Sudden Warmings

2018 **Mexican Geophysical Union Annual Meeting**
28 oct–2 nov 2018 · Puerto Vallarta, Mexico
• Changes in the climatology of major stratospheric sudden warmings using WACCM with high vertical resolution
• Assessment of the ability to reproduce major stratospheric sudden warmings in the CMIP5 historical dataset
• Major stratospheric sudden warmings under climate change scenarios with CMIP5

2017 **European Geosciences Union (EGU) General Assembly 2017**
23–28 abr 2017 · Vienna, Austria
• Estimation of the variability of dust aerosol optical depth over several European regions based on forecast model data

2016 **59th Mexican National Physics Congress (SMF)**
2–7 oct 2016 · León, Mexico
• Stratospheric teleconnection and stratospheric sudden warmings
• Characteristic patterns in the stratospheric variables of the Quasi-Biennial Oscillation
• Stratospheric anomalies before and after stratospheric sudden warmings for the different phases of the Quasi-Biennial Oscillation

2016 **SPARC Workshop SHARP 2016 — Stratospheric Change and its Role for Climate Prediction**
11–15 jul 2016 · Berlin, Germany
• Changes in the stratosphere before and after SSW events in the different phases of the Equatorial QBO
• Differences in the detection and classification of the Stratospheric Sudden Warming over the three reanalyses for the period 1979–2014

2015 **European Geosciences Union (EGU) General Assembly**
12–17 abr 2015 · Vienna, Austria
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings

2015 **SPARC Regional Workshop — Role of the stratosphere in climate variability and prediction**
12–13 ene 2015 · Granada, Spain
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings

2010 **Mexican Geophysical Union Annual Meeting**
7–12 nov 2010 · Puerto Vallarta, Mexico
• Seasonality and anomalies of oceanographic remote-sensing variables in the Gulf of Mexico
• Seasonality and anomalies of sea surface height in the Caribbean Sea using altimetry data
• An alternative for computationally intensive oceanographic data processing through parallel programming in MATLAB

2010 **53rd Mexican National Physics Congress**
25–29 oct 2010 · Boca del Río, Veracruz, Mexico
• High-performance computing applied to parallel programming to optimise processes in Astrophysics and Physical Oceanography

2009 **52nd Mexican National Physics Congress (SMF)**
26–30 oct 2009 · Acapulco, Guerrero, Mexico
• Frequency spectra of meteorological data from María Madre Island
• Comparison of coastal and marine winds in the coastal zone of San Blas, Nayarit (1999–2002)

ACTIVE RESEARCH PROJECTS

2026 – present **Flood early-warning system for the Guadalajara Metropolitan Area — Principal investigator**
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding
Development of an urban flood early-warning system that combines remote sensing (IAM-UdeG weather radar), hydrological runoff modeling over high-resolution LiDAR terrain, and official flood inventories. The system classifies risk by sub-basin in real time and estimates water arrival time, aiming to calibrate thresholds through a retrospective analysis with historical Civil Protection/Fire Department records.

2026 – present **Generative AI for lab practice reports (Google Gems) — Principal investigator**
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding
Implementation of Google Gems as a writing and feedback assistant for students to draft and self-assess their lab practice reports in physics courses.

2025 – present **AI-generated educational comics for science teaching — Principal investigator**
Universidad de Guadalajara · Funding: No external funding
Development and evaluation of educational comics created with generative AI tools as a didactic resource for teaching physics, mathematics and science at basic and upper-secondary levels.

2025 – present **Pedagogical Calculus Assistant with Google Gems: development and student perception — Principal investigator**
Universidad Tecnológica de Jalisco · Funding: Universidad Tecnológica de Jalisco
Design and implementation of a Calculus learning assistant based on Google Gems, with quantitative and qualitative study of students' perception, satisfaction and academic performance.

2025 – 2028 **Optical elements to enhance visual acuity via wavefront coding — Collaborator**
Universidad de Guadalajara · Funding: Universidad de Guadalajara
Design and validation of optical elements that improve visual acuity through wavefront coding.

2025 – present **World Wide Lightning Location Network (WWLLN) — Collaborator · antenna lead**
University of Washington · Funding: University of Washington
Global lightning location network. In charge of the receiving antenna/node and real-time data delivery.

2024 – present **Tlakakini Science & Technology Club — Coordinator**
CUCEI · Universidad de Guadalajara · Funding: University of Guadalajara · CUCEI
Science-outreach club for children, with monthly experiment and workshop sessions.

TECH DEVELOPMENTS

Flood early-warning system · Guadalajara Metro Area
Python · FastAPI · Leaflet · NumPy · pysheds · rasterio · GDAL · Docker
Real-time web platform that anticipates urban flooding in the Guadalajara Metropolitan Area. It fuses the IAM-UdeG weather radar (one scan per minute) with a hydrological runoff model built on 5 m LiDAR and Copernicus DEM terrain, plus the IMEPLAN official flood inventory (363 recurrent sites). It produces a per-zone risk light —calibrated against official water-depth records— and a continuous risk grid that routes rainfall downslope, with concentration time and a "water incoming" ETA. Includes live and historical-replay views. Deployed in production.

CU Tlaquepaque Climatological Forecast System
GFS 0.25° (NOAA/NOMADS) · NASA GIBS/MODIS · WWLLN
Real-time weather forecast web application for Mexico. It displays GFS 0.25° numerical model data (NOAA/NOMADS) and satellite temperature (NASA GIBS/MODIS) on an interactive map with temperature, precipitation, humidity, wind, UV index, air quality and atmospheric electrical activity (WWLLN) layers. It automatically generates a daily climatological bulletin and visual alerts. Work registered with INDAUTOR (no. 03-2026-042712265900-01, 28 Apr 2026; co-authors: C. Almaguer Gómez, J. Durán Curiel) and licensed to CU Tlaquepaque (May 2026) for the benefit of its university community.

Tlakakini web system for science fair management
HTML5 · CSS3 · JavaScript ES6+ · Google Apps Script · Google Sheets
Web application for end-to-end management of the science fairs of the Tlakakini Science and Technology Club (UdeG). It coordinates administrators, workshop leaders (researchers, graduate programs, companies and student chapters) and participants (children and adults): registration and authentication, workshop proposal and approval with schedule and capacity, enrollment with automatic overlap validation, attendance tracking with real-time points and a six-category gamified system, plus an admin dashboard. Serverless architecture. Work registered with INDAUTOR (no. 03-2026-042712282100-01, 28 Apr 2026; co-author: C. Almaguer Gómez) and licensed to CU Tlaquepaque (May 2026).

COMMUNITY BUILDING & SERVICE

Thesis supervision

2011 — **Estacionalidad y anomalías del nivel del mar en el Caribe por medio de datos de altimetría**
Lilia Lizeth Sierra Romero · Bachelor's · Co-advisor · Universidad de Guadalajara

Journal peer review

2026 — **Theoretical and Applied Climatology (Springer Nature)**
Reviewer — international journal

2025 — **IEEE Latin America Transactions**
Reviewer — Q2 journal, double-blind

2024–2025 — **Int. J. of Environment and Pollution (Inderscience)**
Reviewer — Q4 journal, double-blind

2025 — **Congreso de Ingeniería, Ciencia y Gestión Ambiental (AMICA)**
Scientific committee reviewer

Project evaluation

2025–2026 — **SECIHTI**
External evaluator — 2026 National Call, Basic & Frontier Science (3 proposals)

SCIENCE OUTREACH

2024 – — **Tlakakini Science & Technology Club**
Co-founder and coordinator. Monthly outreach sessions for children on national non-school days.

2009 – — **Physics conferences (AGU, EGU, SMF)**
Talks at national and international conferences on climate and the stratosphere.

2005 – — **Training science communicators**
Optics and electromagnetism workshops; National Outreach Meeting.

2009 – — **Red Temática de Medio Ambiente y Sustentabilidad (Environment & Sustainability Thematic Network)**
Member of the CONACYT environment and sustainability network.

2011 – — **Environmental Physics Laboratory (EPhysLab)**
External collaborating researcher with the University of Vigo group.

GRANTS & HONORS

2025–2028 — **National Researcher System · Candidate**
SECIHTI · Mexico

2012–2016 — **FPI Grant · Predoctoral Researcher**
Ministerio de Educación · Spain

2011–2012 — **Climate Science Master's Scholarship**
Campus do Mar · Spain

2011–2012 — **«Ourense Exterior» mobility grant**
Diputación de Ourense · Spain

2009–2011 — **Hydrometeorology Master's Scholarship**
CONACYT · Mexico

2009 **Admission to the Sistema Estatal de Investigación (Jalisco State Research System)**
COECYTJAL · Jalisco, Mexico

2009 **Bachelor's Thesis Scholarship**
SEP · Mexico

CONTINUING EDUCATION

2026 **Course** — Basic English A1: Common verbs · Platzi

2026 **Course** — English vocabulary and expressions · Platzi

2026 **Course** — Basic English A2: Common questions and answers · Platzi

2025 **Diploma** — Techno-pedagogical strategies for developing 21st-century skills · Universidad de Guadalajara

2025 **Course** — Artificial Intelligence for teaching · Universidad de Guadalajara

2025 **Course** — Generative Artificial Intelligence for teachers · Universidad Anáhuac

2025 **Course** — Generative AI in the classroom · UNAM

2025 **Course** — Python for Data Science, AI & Development · IBM

2025 **Certification** — What is Data Science? · IBM

2025 **Course** — Differential Calculus for Data Science and AI · Platzi

2025 **Course** — Statistics and Probability · Platzi

2025 **Course** — Foundations of Software Engineering · Platzi

2025 **Course** — Introduction to the Terminal and Command Line · Platzi

2025 **Course** — Mathematical Functions for Data Science and AI · Platzi

2025 **Course** — Gender equality and prevention of harassment · Universidad de Guadalajara

2025 **Course** — Training for administering and supervising the UdeG admission test (PAA) · Universidad de Guadalajara

2025 **Course** — Computers and Informatics · Platzi

2025 **Course** — Basic Computing · Platzi

2025 **Course** — Fundamentals of Mathematics · Platzi

2025 **Course** — Mathematical Language and Notation · Platzi

2025 **Course** — Basic English A1 for beginners · Platzi

2025 **Course** — Basic English A1: The verb to be · Platzi

2025 **Course** — Basic English A1: Present simple · Platzi

2025 **Course** — Basic English A1: Dates, times and simple expressions · Platzi

2025 **Course** — Strategies for learning English online · Platzi

2025 **Course** — English pronunciation · Platzi

2025 **Course** — English writing · Platzi

2014 **Course** — OOP in Atmospheric and Oceanic Model Development · Universidade de Vigo

2011 **Diploma** — Teaching Competencies: Introduction to General High School Education · Universidad de Guadalajara

2009 **Training** — DOCA Module I: The National System of Technological Education · Universidad de Guadalajara

2005 **Workshop** — Training for Science Communicators · Universidad de Guadalajara

2005 **Course** — Theoretical and Conceptual Development of Electromagnetism (Part I) · Universidad de Guadalajara

LANGUAGES

Spanish — Native · English — Intermediate

TOOLBOX

Python · Fortran · MatLab · IDL · NCL · LaTeX · COMSOL · LabVIEW · MPI · Linux · Moodle · Pandas · Scikit-learn · Jupyter